

U3

Q. Find the radiated resistance of an infinitesimal dipole whose overall length = $\frac{\lambda}{50}$

A. $R_r = \frac{80}{3} \pi \left(\frac{\lambda}{\lambda} \right)^2$

For free space, $\eta = 120\pi \Rightarrow R_r = 80\pi^2 \left(\frac{\lambda}{\lambda} \right)^2 = 80\pi^2 \left(\frac{1/50}{\lambda} \right)^2 = \frac{80\pi^2}{2500} = 0.316 \Omega$

Q. a) A 50W transmitter is radiating its free space using 12dB omnidirectional antenna. The freq. of operation is 900MHz. Calculate Power density & Electric field intensity at a distance of 10km from antenna along the main beam

A. $dB(G_r) = 12 = 10 \log_{10}(G_r)$

$G_r = 15.85$

$W_{av} = \frac{P_t \cdot G_t}{4\pi R^2} = \frac{50 \times 15.85}{4\pi \times (10 \times 10^3)^2} = 6.3065 \times 10^{-7} \text{ W/m}^2$

b) If 1cm long Hertzian dipole is used to receive the signal, what is the open circuit voltage developed at its terminals (Assume R^x is polarization matched with T^x & its pattern max is pointing towards T^x)

A. Open circuit voltage $|V_a| = |E| \cdot l$

Electric field at R^x antenna, $W_{av} = \frac{1}{2} \frac{|E|^2}{\eta} \Rightarrow |E| = \sqrt{\frac{2\eta W_{av}}{1}} = \sqrt{\frac{2 \times 120\pi \times 6.3065 \times 10^{-7}}{1}} = 2.1798 \times 10^{-2} \text{ V/m}$

Magnitude of voltage induced, $|V_a| = |E| \times l$
 $= 2.1798 \times 10^{-2} \times (1 \times 10^{-2})$
 $= 2.1798 \times 10^{-4} \text{ V}$

c) If 100% efficiency, what is power delivered to its matched load.

A. $R_r = 80\pi^2 \left(\frac{\lambda}{\lambda} \right)^2$
 $= 80\pi^2 \left(\frac{\lambda}{c/f} \right)^2 = 80\pi^2 \times \left(\frac{10^{-2}}{\frac{3 \times 10^8}{900 \times 10^6}} \right)^2 = 720\pi^2 \times 10^{-9} = 0.709 \Omega$

$P_{rad} = \frac{|V_a|^2}{8 R_{rad}} = \frac{(2.1798 \times 10^{-4})^2}{8 \times 0.709} = 8.36 \times 10^{-9} \text{ W}$

d) Verify that the friss T^x formulae also gives same values as obtained in Q.c)

A. $P_r = P_t \cdot G_t \cdot G_r \cdot \left(\frac{\lambda}{4\pi R} \right)^2$
 $= 50 \times 15.85 \times 1.5 \times \left(\frac{1/3}{4\pi \times 10 \times 10^3} \right)^2 = 8.36 \times 10^{-9} \text{ W}$

Q. Calculate the radiation efficiency of an antenna if i/p power is 100W and power dissipated in it is 1W

A. $P_{in} = 100 \text{ W}$

$P_{rad} = P_{in} - P_{out} = 100 - 1 = 99 \text{ W}$

$\rho_r = \frac{P_{rad}}{P_{in}} = \frac{99}{100}$

Q. Determine the fields due to Hertzian dipole placed at origin, if the maximum power density at a distance of 100m from the dipole is 0.2 mW/m^2 in free space

A. $w_{av} = \frac{\eta}{2} |H\phi|^2$

$$0.2 \times 10^{-3} = \frac{120\pi}{2} |H\phi|^2 \Rightarrow |H\phi| = 1.03 \times 10^{-3} \text{ A/m}$$

$$H\phi = j 1.030 \times 10^{-3} e^{jk_{100}} \text{ A/m}$$

$$E_{\theta} = \eta H\phi = j 120\pi \times 1.030 \times 10^{-3} e^{jk_{100}} \text{ A}$$

Q. Determine rms current in Hertzian dipole of length $\lambda/50$, radiating a power of 5W in free space

A. $I(t) = \text{Re} \{ I_0 e^{j\omega t} \} = I_0 \cos \omega t$

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

$$\begin{aligned} P_{rad} &= 10 \kappa^2 I_0^2 \lambda^2 \\ &= 10 \omega^2 \mu \epsilon I_0^2 \lambda^2 \\ &= 10 \times \left(\frac{2\pi c}{\lambda} \right)^2 \times \frac{1}{c^2} \times I_0^2 \times \left(\frac{\lambda}{50} \right)^2 \end{aligned}$$

$$5 = \frac{4\pi^2 I_0^2}{2500} \Rightarrow I_0^2 = 31.66 \Rightarrow I_0 = 5.627 \text{ A}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 3.978 \text{ A}$$

Q. Determine the power radiated by a dipole of length $\lambda/50$ which carries a current of 5A

A. $c = \lambda f$

$$f = \frac{c}{\lambda}$$

$$\omega = 2\pi f = \frac{2\pi c}{\lambda} \quad \& \quad \eta = 120\pi$$

$$\kappa = \omega \sqrt{\mu \epsilon} = \frac{2\pi}{\lambda}$$

$$P = \eta \left(\frac{\kappa^2 I^2 \lambda^2}{12\pi} \right) = 10 \times \left(\frac{2\pi}{\lambda} \right)^2 \cdot 5^2 \cdot \left(\frac{\lambda}{50} \right)^2 = \frac{10 \times 4\pi^2 \times 25}{2500} = \frac{4\pi^2}{10} = 3.95 \text{ W}$$

Q. A Hertzian dipole of $l = 5 \text{ cm}$ is radiating into free space at 100 MHz . Determine the power radiated & the radiation resistance of the dipole if the instantaneous current flowing into the dipole is $i(t) = 10 \cos(2\pi \times 10^8 t)$

A. $\omega = 2 \times 10^8 \text{ rad/s}$

$$l = 5 \text{ cm}$$

$$f = 10^8 \text{ Hz}$$

$$\lambda = \frac{c}{f} = 3 \text{ m}$$

$$R_{rad} = 80 \pi^2 \left(\frac{l}{\lambda} \right)^2 = 80 \pi^2 \left(\frac{0.05}{3} \right)^2 = 0.219 \Omega$$

$$P = \frac{I^2 R_{rad}}{2} = \frac{(10)^2}{2} \times 0.219 = 10.95 \text{ W}$$

Q. A short dipole of 15cm is operating at 200MHz. Determine the power radiated & radiation resistance of the dipole if the instantaneous current flowing in the dipole is $i(t) = 5\cos(\omega t)$

A. $\text{Re}\{I e^{j\omega t}\} = I \cos(\omega t)$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{200 \times 10^6} = 1.5\text{m}$$

$$R_{\text{rad}} = 20\pi^2 \left(\frac{l}{\lambda}\right)^2 = 20\pi^2 \left(\frac{0.15}{1.5}\right)^2 = 1.974 \Omega$$

$$P_{\text{rad}} = \frac{I^2 R_{\text{rad}}}{2} = \frac{5^2 \times 1.974}{2} = 24.67\text{W}$$

Q. A short dipole radiates 10W of power in space. Determine max magnitude of Electric field at a distance of 30m

A. $|E_{\theta}|_{\text{max}} = \frac{\kappa^2 I l}{8\pi\omega\epsilon r} \Rightarrow I l = \frac{8\pi\omega\epsilon r}{\kappa^2} \cdot |E_{\theta}|_{\text{max}}$

$$P_{\text{rad}} = \frac{\eta \kappa^2 I^2 l^2}{48\pi} = \frac{\eta \kappa^2}{48\pi} \times \frac{64\pi^2 \omega^2 \epsilon^2 r^2}{\kappa^4 \kappa^2} |E_{\theta}|_{\text{max}}^2$$

$$= \frac{10}{\frac{120\pi}{\kappa}} \times \frac{16}{64\pi^2} \times \frac{10^8 \epsilon^2 r^2}{\omega^2 \mu \epsilon} \cdot |E_{\theta}|_{\text{max}}^2$$

$$= 160\pi^2 \cdot \frac{\epsilon}{\mu} \cdot (30)^2 \cdot |E_{\theta}|_{\text{max}}^2$$

$$= \frac{160\pi^2 \times 900}{\frac{\mu}{\epsilon}} |E_{\theta}|_{\text{max}}^2$$

$$= \frac{160\pi^2 \times 900}{120\pi \times 120\pi} \cdot |E_{\theta}|_{\text{max}}^2$$

$$= 10 |E_{\theta}|_{\text{max}}^2 = 10 \Rightarrow |E_{\theta}|_{\text{max}} = 1 \text{ V/m}$$

Q. A short dipole of length 0.1λ is kept symmetrical at origin oriented along z-direction & radiating 1KW power into free space. Calculate power density at a distance of 1Km along the direction $\theta = 45^\circ$ & 90°

A. $W_{\text{av}} = \frac{\kappa^3 (I l)^2 \sin^2 \theta}{128\pi^2 r^2 \omega \epsilon}$

$$P_{\text{rad}} = \frac{\eta \kappa^2 (I l)^2}{48\pi} \Rightarrow (I l)^2 = \frac{P_{\text{rad}} \times 48\pi}{\eta \kappa^2}$$

$$W_{\text{av}} = \frac{\kappa^3}{128\pi^2 r^2 \omega \epsilon} \times \frac{P_{\text{rad}} \times 48\pi}{\eta \kappa^2} \times \frac{1}{2}$$

$$= \frac{10 \sqrt{\mu \epsilon} \times 10^3}{r^2 \sqrt{\epsilon} \eta} \times \frac{48\pi}{128\pi^2 \times 2}$$

$$= \frac{10^3}{(10^3)^2} \times \frac{\sqrt{\mu}}{\sqrt{\epsilon}} \times \frac{1}{\eta} \times \frac{0.1875}{\pi}$$

$$= 59.68 \times 10^{-6} \text{ W/m}^2$$

$$\left(\eta = \sqrt{\frac{\mu}{\epsilon}} \right)$$

Q. Calculate the max power density at a distance of 1000m from a Hertzian dipole radiating 10W of power. Assume that dipole has no losses. What are E & H field intensities at the given point? If the same power were to be radiated by an isotropic antenna, what will be power density & field intensities? How much power should the isotropic antenna radiate so the field antennas of isotropic & Hertzian dipole are same?

A. $P_{rad} = 10W$; $r = 1000m$

$$D = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi W_{av} r^2}{P_{rad}} = 1.5$$

$$W_{av} = \frac{1.5 P_{rad}}{4\pi r^2} = \frac{1.5 \times 10}{4\pi \times 10^6} = 1.1937 \times 10^{-6} W/m^2$$

$$W_{av} = \frac{|E|^2}{\eta} \Rightarrow |E| = \sqrt{\eta W_{av}} = \sqrt{120\pi \times 1.1937 \times 10^{-6}} = 0.0212 V/m$$

$$|H| = \frac{|E|}{\eta} = \frac{0.0212}{120\pi} = 5.627 \times 10^{-5} A/m$$

$$W_{av, max} = \frac{P_{rad}}{4\pi r^2}$$

For max fields of Hertzian dipole & isotropic antenna to be identical

$$W_{av} = W_{av, max}$$

$$\begin{aligned} P_{rad} &= 4\pi r^2 \times W_{av, max} \\ &= 4\pi \times 10^6 \times 1.1937 \times 10^{-6} \\ &= 15W \end{aligned}$$

Q. Calculate beamwidth in XY & YZ plane of an antenna whose radiation intensity is $U(\theta, \phi) = \begin{cases} \sin^2\theta \sin\phi, & 0 < \phi < \pi \\ 0 & 0 < \theta < \pi \end{cases}$, otherwise

A. In XY plane, $\theta = \pi/2$

$$U(\pi/2, \phi) = \sin^2(\pi/2) \sin\phi = \sin\phi$$

$$\text{For HPBW, } U(\pi/2, \phi) = 0.5 = \sin\phi$$

$$\phi = 30^\circ \text{ or } 150^\circ$$

$$\text{Hence HPBW in XY plane} \Rightarrow \phi_n = 150 - 30 = 120^\circ$$

In YZ plane, $\phi = \pi/2$

$$U(\theta, \pi/2) = \sin^2\theta \sin\pi/2 = \sin^2\theta$$

$$\text{For HPBW, } U(\theta, \pi/2) = 0.5 = \sin^2\theta \Rightarrow \sin\theta = 1/\sqrt{2}$$

$$\theta = 45^\circ \text{ or } 135^\circ$$

$$\text{Hence HPBW in YZ plane} \Rightarrow \phi_n = 135 - 45 = 90^\circ$$

Q. An antenna is having a field pattern given by $E(\theta) = \cos\theta$ for $0 < \theta < 90^\circ$. Find HPBW

A. $P \propto E^2 \Rightarrow E \propto \sqrt{P}$

$$\text{So at HPBW, } P = 0.5 \text{ \& } E = 0.707$$

$$\text{So max E field} \Rightarrow E(\theta) = \cos\theta = 1/\sqrt{2}$$

$$\theta = 45^\circ \text{ or } 135^\circ$$

$$\theta_n = 135 - 45 = 90^\circ$$

Q. Calculate directivity of an antenna whose radiation intensity is $U(\theta, \phi) = \begin{cases} \sin\theta \cdot \sin\phi, & 0 < \theta < \pi \\ 0, & \text{otherwise} \end{cases}$

A. $P_{rad} = \int_S W_{av} \cdot da$

$$da = r^2 \sin\theta \, d\theta \, d\phi$$

$$U = r^2 W_{av} \Rightarrow W_{av} = U/r^2$$

$$P_{rad} = \int_S \left(\frac{U}{r^2} \right) \cdot r^2 \sin\theta \, d\theta \, d\phi$$

$$= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} U \cdot \sin\theta \, d\theta \, d\phi$$

$$= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (\sin\theta \sin\phi) \sin\theta \, d\theta \, d\phi = \pi \text{ Watts}$$

$$D = \frac{4\pi U}{P_{rad}} = \frac{4\pi \sin\theta \sin\phi}{\pi} = 4 \sin\theta \sin\phi$$

$$\text{Max directivity } D_{max} = 4 \text{ @ } \theta = \frac{\pi}{2} \text{ \& } \phi = \frac{\pi}{2}$$

$$D_{max} \text{ (dB)} = 10 \log_{10}(4) = 6.02 \text{ dB}$$

Q. Determine directivity of a Hertzian dipole

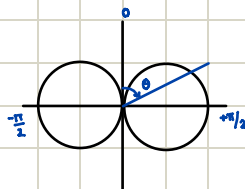
A. $D = \frac{4\pi U}{P_{rad}}$

$$P_{rad} = \frac{\eta k^2 I^2 l^2}{12\pi}$$

$$U = \frac{\eta k^2 I^2 l^2}{32\pi^2} \cdot \sin^2\theta$$

$$D = \frac{4\pi \cdot \sin^2\theta \cdot 12\pi}{32\pi^2} = 1.5 \sin^2\theta$$

$$D_{max} = 1.5 \text{ at } \theta = \pm \frac{\pi}{2}$$



Q. Find the radiation resistance of Hertzian dipole of length $\lambda/60$ & $\lambda/80$

A. $R_{rad} = \frac{80\pi^2 l^2}{\lambda^2}$

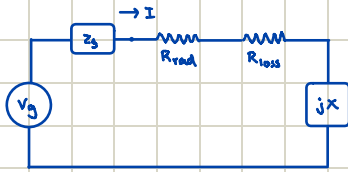
1) $l = \lambda/60$

$$R_{rad} = \frac{80 \times \pi^2 \times \left(\frac{\lambda}{60}\right)^2}{\lambda^2} = \frac{80\pi^2}{60^2} = 0.128$$

2) $l = \lambda/80$

$$R_{rad} = \frac{80 \times \pi^2 \times \left(\frac{\lambda}{80}\right)^2}{\lambda^2} = \frac{80\pi^2}{80^2} = 0.123$$

Q. A Voltage source $V_g = 50 + j40$ Volts, with source impedance $Z_g = 50 \Omega$ is connected to an antenna with $R_{rad} = 70 \Omega$ & $R_{loss} = 1 \Omega$ and reactance of $j25 \Omega$. Calculate the radiation/antenna efficiency, the real power delivered by source, the real input to antenna, power radiated by the antenna, and the power dissipated in the antenna.



$$A. \quad I = \frac{V_g}{Z_g + R_{rad} + R_{loss} + jX} = \frac{50 + j40}{50 + 70 + 1 + j25} = 0.518 e^{j27^\circ} \text{ Amp}$$

$$P_g = \frac{1}{2} \operatorname{Re} \{ V_g I^* \}$$

$$I^* = 0.518 e^{-j27^\circ}$$

$$V_g = 50 + j40 = 64.03 e^{j38.65^\circ}$$

$$P_g = \frac{1}{2} \times \operatorname{Re} \left\{ 0.518 \times e^{-j27^\circ} \times 64.03 \times e^{j38.65^\circ} \right\} = 16.24 \text{ W}$$

$$\text{Real power } P_a = \frac{1}{2} \operatorname{Re} [V_a I^*] = \frac{1}{2} \operatorname{Re} [(70 + 1) I \cdot I^*] = \frac{1}{2} \times 71 \times |I|^2 = \frac{1}{2} \times 71 \times 0.518^2 = 9.525 \text{ Watts}$$

$$\text{Power radiated } P_{rad} = \frac{1}{2} \operatorname{Re} [V_e I^*] = \frac{1}{2} \operatorname{Re} [70 \times I \cdot I^*] = \frac{1}{2} \times 70 \times |I|^2 = \frac{1}{2} \times 70 \times 0.518^2 = 9.39 \text{ Watts}$$

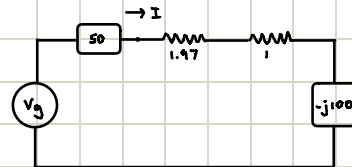
$$\text{Power dissipated } P_{loss} = \frac{1}{2} \operatorname{Re} [V_{loss} I^*] = \frac{1}{2} \operatorname{Re} [1 \times I \cdot I^*] = \frac{1}{2} \times 1 \times |I|^2 = \frac{1}{2} \times 1 \times 0.518^2 = 0.134 \text{ Watts}$$

$$\text{Antenna efficiency } (e) = \frac{P_{rad}}{P_a} = \frac{70}{71} = 0.986 = 98.6\%$$

Q. Calculate R_{rad} & e of Hertzian dipole of length $dl = 0.05\lambda$ having loss resistance of 1Ω and resistance of $-j100 \Omega$. If dipole is connected to a source of $100V$ (peak) having impedance of 50Ω , Calculate real power radiated by antenna & power delivered by the source.

$$A. \quad R_{rad} = 80\pi^2 \frac{l^2}{\lambda^2} = \frac{80\pi^2 \cdot (0.05\lambda)^2}{\lambda^2} = 1.97 \Omega$$

$$e = \frac{R_{rad}}{R_{rad} + R_{loss}} = \frac{1.97}{1.97 + 1} = 0.66$$



$$\text{Real power radiated by antenna, } P_{rad} = \frac{1}{2} \operatorname{Re} [V I^*] = \frac{1}{2} \operatorname{Re} [R_{rad} \times I \times I^*] = \frac{1}{2} R_{rad} \times |I|^2$$

$$I = \frac{V_g}{R_{rad} + R_{loss} + Z_g + jX} = \frac{100}{1.97 + 1 + 50 - j100} = 0.88 \angle 0.62$$

$$P_{rad} = \frac{1}{2} \times 1.97 \times |0.88|^2 = 0.77$$

$$\text{Power delivered by source, } P_g = \frac{1}{2} \operatorname{Re} [V_g I^*] = \frac{1}{2} \operatorname{Re} [(52.97 - j100) \times I \times I^*] = \frac{|I|^2}{2} \operatorname{Re} (52.97 - j100) = 20.68 \text{ W}$$

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Q. Let Power received be -20 dBm , Calculate The power in Watts

A. $-20 = 10 \log_{10} \frac{P_r}{1 \times 10^{-3}}$

$$-2 = \log_{10} \frac{P_r}{10^{-3}} \Rightarrow 10^{-2} = \frac{P_r}{10^{-3}} \Rightarrow P_r = 10^{-5} \text{ Watts}$$

Q. 2 antennas (one Tx & one Rx) are placed facing each other 5 km distance apart. Tx antenna is radiating power of 100W. If the antennas are $\lambda/2$ dipoles & operating freq. is 300 MHz. Determine power received in Watts & in dB

A. $r = 5 \text{ km}$ $f = 300 \text{ MHz}$

$P_t = 100 \text{ W}$ $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ m}$

$G = G_t = G_r = D$

$D_{\max} \left(\frac{\lambda}{2} \right) = 1.642$ @ $\theta = \frac{\pi}{2}$

Pathloss = $20 \log_{10} \left(\frac{\lambda}{4\pi r} \right) = -95.9636$

$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi r} \right)^2 = 6.83 \times 10^{-8} \text{ W}$

$P_r (\text{dB}) = 10 \log_{10} \left(\frac{P_r}{10^{-3}} \right) = -41.65 \text{ dB}$

Example: A 50 W transmitter is radiating into free space using 12 dB omnidirectional antenna. The frequency of operation is 900 MHz.

a) Calculate power density and electric field intensity at a distance of 10 km from the antenna, along the main beam.

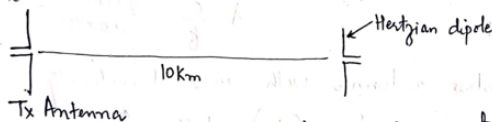
b) If 1 cm long Hertzian dipole is used to receive the signal, what is the open-circuit voltage developed at its terminals. (Assume that Rx is polarization matched with the Tx and its pattern-max is pointing towards Tx).

c) If the Hertzian dipole has 100% efficiency, what is power delivered to its matched load (at the receiver).

d) Verify that the Friis transmission formula also gives same values as obtained in (c).

Sol: Gain of Tx antenna $G_t \text{ dB} = 12 = 10 \log_{10}(G_t)$
So, $G_t = 15.85$

$G_t \text{ dB} = 12$



(a) The Power density close to receiver antenna (due to power transmitted by Tx antenna)

$W_{\text{avt}} = \frac{P_t G_t}{4\pi r^2}$

$W_{\text{avt}} = \frac{50 \times 15.85}{4\pi \times (10,000)^2} = 6.3065 \times 10^{-7} \text{ Watts/m}^2$

Electric field (magnitude) at the receiver antenna

$W_{\text{avt}} = \frac{1}{2} \frac{|E|^2}{\eta}$ Hence, $|E| = \sqrt{2\eta W_{\text{avt}}}$

$|E| = \sqrt{2 \times 120\pi \times 6.3065 \times 10^{-7}} \Rightarrow |E| = 2.179 \times 10^{-2} \text{ V/m}$

b) Magnitude of voltage induced at the receiver antenna

$|V_a| = |E| dl = 2.179 \times 10^{-2} \times 10^{-2} = 2.179 \times 10^{-4} \text{ Volts}$

Radiation resistance of Hertzian dipole (receiver antenna)

$R_{\text{rad}} = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2$; $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = \frac{1}{3} \text{ m}$

$R_{\text{rad}} = 80\pi^2 \left(\frac{10^{-2} \times 3}{1} \right)^2$ Hence $R_{\text{rad}} = 0.71 \Omega$

(c) Power delivered to the matched load (at the receiver).

$P_r = \frac{|V_a|^2}{8 R_{\text{rad}}} = \frac{(2.179 \times 10^{-4})^2}{8 \times 0.71}$

$P_r = 8.36 \times 10^{-9} \text{ Watts}$

(d) Friis transmission formula.

Power delivered to the receiver load

$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi r} \right)^2 = 50 \times 15.85 \times 1.5 \left(\frac{1}{3 \times 4\pi \times 10^4} \right)^2$

$P_r = 8.36 \times 10^{-9} \text{ Watts}$

15 for Hertzian dipole

Q. Plot AF (normalized) for $d = 0.25\lambda$

A. $|AF|_{\text{nor}} = \left| \cos \left(\frac{kd}{2} \cos \theta \right) \right|$

$$d = 0.25\lambda \Rightarrow \frac{kd}{2} = \frac{2\pi}{\lambda} \times \frac{0.25\lambda}{2} = \frac{\pi}{4}$$

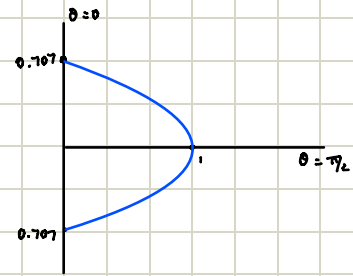
$$|AF|_{\text{nor}} = \left| \cos \left(\frac{\pi}{4} \cos \theta \right) \right|$$

$$m \leq \frac{d}{\lambda} = \frac{0.25\lambda}{\lambda} \Rightarrow m \leq 0.25 \quad (m \text{ is integer \& only } m=0 \text{ satisfies})$$

$$\cos \theta_m = 0 \Rightarrow \theta_m = \pi/2 \quad (\text{maxima at } \pi/2)$$

$$\text{when } \theta = 0, \quad \left| \cos \left(\frac{\pi}{4} \times 1 \right) \right| = 0.707$$

$$\text{when } \theta = \pi \quad \left| \cos \left(\frac{\pi}{4} \times -1 \right) \right| = 0.707$$



$$n \leq \frac{d}{\lambda} + \frac{1}{2} \Rightarrow n \leq \frac{0.25\lambda}{\lambda} + \frac{1}{2} \Rightarrow n \leq 0.75$$

↓
 $n=0$
↓
No nulls

Q. Plot AF (normalized) for $d = 0.5\lambda$

A. $|AF|_{\text{nor}} = \left| \cos \left(\frac{kd}{2} \cos \theta \right) \right|$

$$d = 0.5\lambda \Rightarrow \frac{kd}{2} = \frac{2\pi}{\lambda} \times \frac{0.5\lambda}{2} = \frac{\pi}{2}$$

$$|AF|_{\text{nor}} = \left| \cos \left(\frac{\pi}{2} \cos \theta \right) \right|$$

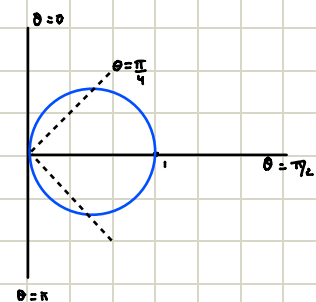
$$m \leq \frac{d}{\lambda} = \frac{0.5\lambda}{\lambda} \Rightarrow m \leq 0.5 \quad (m \text{ is integer \& only } m=0 \text{ satisfies})$$

$$\cos \theta_m = 0 \Rightarrow \theta_m = \pi/2 \quad (\text{maxima at } \pi/2)$$

$$\text{when } \theta = 0, \quad \left| \cos \left(\frac{\pi}{2} \times 1 \right) \right| = 0$$

$\Rightarrow 2 \text{ Nulls}$

$$\text{when } \theta = \pi \quad \left| \cos \left(\frac{\pi}{2} \times -1 \right) \right| = 0$$



$$n \leq \frac{d}{\lambda} + \frac{1}{2} \Rightarrow n \leq \frac{0.5}{1} + \frac{1}{2} = 1$$

$$\cos \theta_n = \pm \left(\frac{2n-1}{2} \right) \left(\frac{\lambda}{d} \right) = \pm \frac{1}{2} \cdot \frac{\lambda}{0.5\lambda} = \pm 1$$

$$\theta_n = \cos^{-1}(\pm 1) = 0, 180^\circ$$

Q. Plot AF (normalized) for $d = \lambda$

A. $|AF|_{\text{nor}} = \left| \cos \left(\frac{kd}{2} \cos \theta \right) \right|$

$$d = \lambda \Rightarrow \frac{kd}{2} = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

$$|AF|_{\text{nor}} = \left| \cos \left(\pi \cos \theta \right) \right|$$

$$m \leq \frac{d}{\lambda} = \frac{\lambda}{\lambda} \Rightarrow m \leq 1 \quad (m \text{ is integer \& only } m=0 \& 1 \text{ satisfies})$$

$$\cos \theta_m = \pm \frac{m\lambda}{d}$$

When $m=0$,

$$\cos \theta_m = 0$$

$$\theta_m = \cos^{-1}(0)$$

$$= \pi/2$$

$$\theta_m = 0, \pi/2, \pi \Rightarrow 3 \text{ maxima}$$

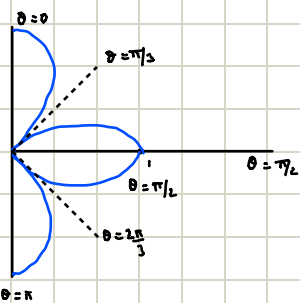
2 minima

When $m=1$

$$\cos \theta_m = \frac{\lambda}{\lambda} = \frac{\lambda}{\lambda} = 1$$

$$\theta_m = \cos^{-1}(1)$$

$$= 0 \& \pi$$



$$n \leq \frac{d}{\lambda} + \frac{1}{2} = \frac{\lambda}{\lambda} + 0.5 = 1.5 \Rightarrow n=1$$

$$\cos \theta_n = \pm \left(\frac{2n-1}{2} \right) \left(\frac{\lambda}{d} \right)$$

$$= \pm \frac{1}{2} \cdot 1 = \pm 0.5$$

$$\theta_n = \cos^{-1}(\pm 0.5) = \frac{\pi}{3} \& \frac{2\pi}{3}$$

Q. Plot the $|AF|_n$ for $d = 0.75\lambda$ for 2-element end-fire array ($\alpha = \pi d$)

A. $d = 0.75\lambda$

For maxima,

$$0 \leq m \leq \frac{2d}{\lambda}$$

$$0 \leq m \leq \frac{2}{\lambda} \times \frac{3\lambda}{4} \Rightarrow 0 \leq m \leq \frac{3}{2} \Rightarrow m = 0, 1$$

For $m = 0$,

$$\cos \theta_m = \frac{m\lambda}{d} - 1 = -1 \Rightarrow \theta_m = \pi$$

For $m = 1$,

$$\cos \theta_m = \frac{m\lambda}{d} - 1 = \frac{\lambda}{\frac{3\lambda}{4}} - 1 = \frac{4}{3} - 1 = \frac{1}{3} \Rightarrow \theta_m = 70.5^\circ$$

For nulls,

$$n \leq \frac{2d}{\lambda} + \frac{1}{2}$$

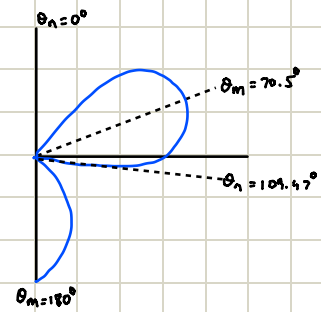
$$n \leq \frac{2 \times \frac{3\lambda}{4}}{\lambda} + \frac{1}{2} \Rightarrow n \leq 2 \Rightarrow n = 1, 2$$

For $n = 1$,

$$\cos \theta_n = \frac{2n-1}{2} \cdot \frac{\lambda}{d} - 1 = \frac{1}{2} \cdot \frac{4}{3} - 1 = -\frac{1}{3} \Rightarrow \theta_n = 109.47^\circ$$

For $n = 2$

$$\cos \theta_n = \frac{2n-1}{2} \cdot \frac{\lambda}{d} - 1 = \frac{3}{2} \cdot \frac{4}{3} - 1 = 1 \Rightarrow \theta_n = 0^\circ$$



Example: For $d = 0.25\lambda$ Plot the $|AF|_n$ for two-element end-fire array.

Sol:-

For $d = 0.25\lambda$

For maxima,

$$0 \leq m \leq \frac{2d}{\lambda}$$

$$0 \leq m \leq \frac{2 \times \lambda}{4 \times \lambda}$$

$$0 \leq m \leq 0.5$$

$$m = 0$$

For $m = 0$

$$\cos \theta_m = \frac{m\lambda}{d} - 1$$

$$\cos \theta_m = -1$$

$$\theta_m = \cos^{-1}(-1)$$

$$\theta_m = \pi$$

For Null

$d = 0.25\lambda$

$$n \leq \frac{2d}{\lambda} + \frac{1}{2}$$

$$n \leq \frac{2 \times \lambda}{4 \times \lambda} + \frac{1}{2}$$

$$n \leq 1$$

$$n = 1$$

$n = 1$

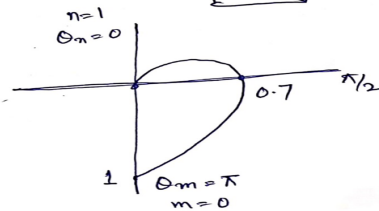
$$\cos \theta_n = \left(\frac{2n-1}{2} \right) \frac{\lambda}{d} - 1$$

$$= \left(\frac{1}{2} \right) \left(\frac{\lambda}{\lambda} \right) - 1$$

$$\cos \theta_n = 1$$

$$\theta_n = \cos^{-1}(1)$$

$$\theta_n = 0$$



For $\theta = \pi/2$

$$|AF|_n = \cos \left[\frac{k d}{2} (1 + \cos \theta) \right]$$

$$= \cos \left[\frac{2\pi \times \lambda}{4 \times \lambda} (1 + \cos \pi/2) \right]$$

$$|AF|_n = \cos(\pi/4) = 0.707$$

Q. 2 identical & 11^th antennas are kept apart by distance of $\lambda/4$ in the H-plane. They are fed with equal currents but rightmost antenna has phase shift of $+90^\circ$. Draw the radiation pattern for the normalized array pattern

A. $N = 2$, Uniform Antenna

$$I_1 = I_0 e^{j(0)}$$

$$I_2 = I_0 e^{j(\pi/2)}$$

$$\Psi = kd \cos \theta + \alpha$$

$$\alpha = \pi/2$$

$$d = \lambda/4$$

$$\Psi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} \cdot (\cos \theta) + \frac{\pi}{2}$$

$$= \frac{\pi}{2} (1 + \cos \theta)$$

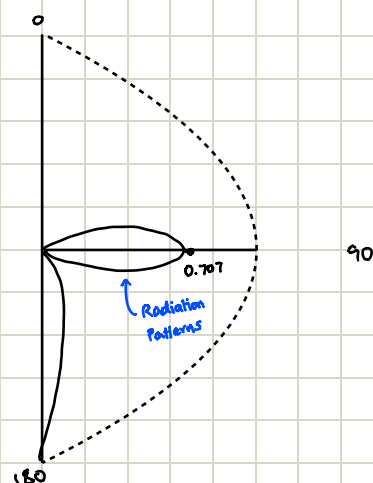
$$AF = \frac{\sin \frac{N\Psi}{2}}{N \sin \frac{\Psi}{2}} = \frac{\sin \Psi}{2 \sin \frac{\Psi}{2}} = \frac{2 \sin \frac{\Psi}{2} \cdot \cos \frac{\Psi}{2}}{2 \sin \frac{\Psi}{2}} = \cos \frac{\Psi}{2}$$

$$= \cos \left(\frac{\pi}{4} (1 + \cos \theta) \right)$$

$$\text{When } \theta = 0^\circ \Rightarrow AF = 0$$

$$\theta = 90^\circ \Rightarrow AF = \frac{1}{\sqrt{2}}$$

$$\theta = 180^\circ \Rightarrow AF = 1$$



Q. 2 halfwave dipole antennas are placed as shown in the figure are excited with sinusoidally varying freq of 3 MHz and phase shift of $\pi/2$ between them. The element at the origin leads in phase. If the maximum radiated E field at a point P in xy plane occurs in an azimuthal angle of 60° , then distance b/w antennas = ?

A. $\Psi = kd \cos \theta + \alpha$

$$\alpha = -\pi/2$$

$$\Psi = \frac{2\pi}{\lambda} d \cos 60^\circ - \frac{\pi}{2}$$

$$= \frac{\pi d}{\lambda} - \frac{\pi}{2}$$

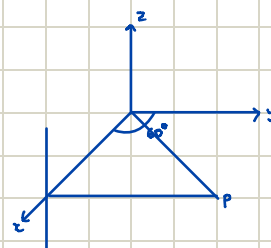
E-field max @ $\Psi = 0$

$$\frac{\pi d}{\lambda} - \frac{\pi}{2} = 0$$

$$\frac{\pi d}{\lambda} = \frac{\pi}{2} \Rightarrow \frac{d}{\lambda} = \frac{1}{2} \Rightarrow \lambda = 2d$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^6} = 100 \text{ m}$$

$$d = \frac{\lambda}{2} = 50 \text{ m}$$



Q. Plot the array factor for $d = 0.75\lambda$ for a 2 element endfire array

$$A. \quad 0 \leq m \leq \frac{2d}{\lambda} \Rightarrow 0 \leq m \leq \frac{2}{\lambda} \cdot \frac{3\lambda}{4} \Rightarrow 0 \leq m \leq 1.5$$

$$m = 0, 1$$

$$\theta_m = \cos^{-1} \left(\pm \left(\frac{m\lambda}{d} - 1 \right) \right)$$

$$m = 0, \quad \theta_m = \cos^{-1}(-1) = 180^\circ$$

$$m = 1, \quad \theta_m = \cos^{-1}(1/3) = 70.5^\circ$$

$$n \leq \left(\frac{2d}{\lambda} + \frac{1}{2} \right) = 2$$

$$n = 1, 2$$

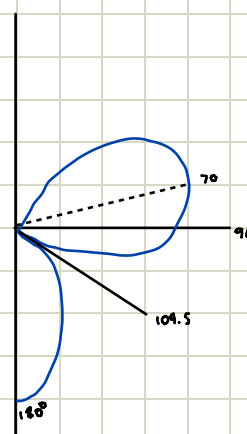
$$\theta_n = \cos^{-1} \left(\left(\frac{2n-1}{2} \right) \frac{\lambda}{d} - 1 \right)$$

$$n = 1$$

$$\theta_n = \cos^{-1} \left(-\frac{1}{3} \right) = 109.5^\circ$$

$$n = 2$$

$$\theta_n = \cos^{-1}(1) = 0^\circ$$

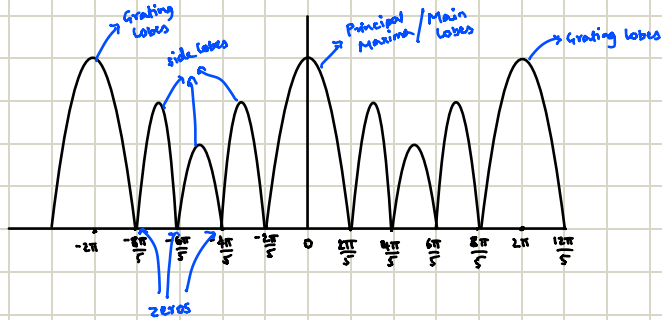


Q. For a 5 element uniform array plot the normalized AF.

A. For N element, N-1 zeros & N-2 side lobes

$$\Psi = \pm \frac{2\pi}{N} = \pm \frac{2\pi}{5}, \pm \frac{4\pi}{5}, \pm \frac{6\pi}{5}, \pm \frac{8\pi}{5}, \pm \frac{10\pi}{5}$$

Dont take $p=0 \& N$. Maximas at $\pm 2m\pi = 0, \pm 2\pi, \pm 4\pi, \dots$



Q. Determine 3dB beamwidth of 7 element array having phase shift $\alpha=0$ & element separation of 0.5λ

Hint: Ψ lies b/w 22 & 24

A. $N=7$ $\alpha=0$ $\Psi = 22^\circ$ to 24°

$$AF = \sin \frac{7\Psi}{2} = 0.707 \quad (\text{ cuz } 3\text{dB})$$

$$\frac{7 \sin \frac{\Psi}{2}}$$

$$\Psi = 23 = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} \cdot \cos \theta \Rightarrow 23 = \pi \cos \theta$$

$$\theta_1 = 82.7^\circ \quad \theta_2 = 97.34^\circ$$

$$\text{HPBW} = 97.34 - 82.7 = 14.64$$

Q. Determine 3dB beamwidth of 5 element uniform array & 5 element binomial array having phase shift $\alpha=0$ & element separation of 0.5λ (Hint: Ψ lies b/w 32 & 34)

A.

$$|AF|_n = \frac{\sin(\frac{5\Psi}{2})}{5 \sin \frac{\Psi}{2}} = 0.707 \Rightarrow \sin(\frac{5\Psi}{2}) - 3.535 \sin \frac{\Psi}{2} = 0$$

If $\Psi = 32 \Rightarrow \text{diff} = 0.104$

$\Psi = 32.5 \Rightarrow \text{diff} \approx 0$ ✓

$\Psi = 33 \Rightarrow \text{diff} = -0.01$

$\Psi = 33.5 \Rightarrow \text{diff} = -0.027$

$\Psi = 34 \Rightarrow \text{diff} = -0.037$

$$\Psi = \pm 32.5$$

$$\Psi = \alpha + Kd \cos \theta$$

$$= 0 + \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} \cdot \cos \theta$$

$$\pm 32.5 = 180 \times \cos \theta$$

$$\theta = \cos^{-1} \left(\pm \frac{32.5}{180} \right) = 79.59^\circ \text{ or } 100.4^\circ$$

$$3\text{dB beamwidth} = 100.4^\circ - 79.59^\circ = 20.81^\circ$$

To plot θ pattern,

→ Nulls occur at $\Psi_n = \pm 2\pi \left(\frac{n}{N} \right)$, $n=1,2,3,\dots$

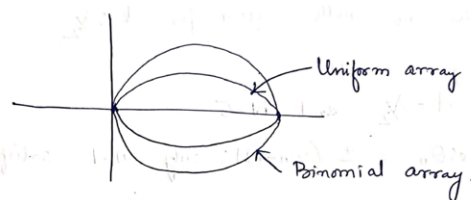
$$\Psi_1 = \pm \frac{2\pi}{N} = \pi \cos \theta_n$$

$$\pm \frac{2\pi}{5} = \pi \cos \theta_n \Rightarrow \theta_n = 66.42^\circ \text{ or } 113.57^\circ$$

$$\Psi_2 = \pm \frac{4\pi}{5} = \pi \cos \theta_n$$

$$\pm \frac{4\pi}{5} = \pi \cos \theta_n \Rightarrow \theta_n = 36.87^\circ \text{ or } 143.13^\circ$$

$$AF @ 0^\circ = 1$$



Binomial Array

$$\begin{aligned} AF &= I_0 (1+z_0)^{S-1} = I_0 (1+z_0)^N = I_0 |(1+e^{j\Psi})^N| \\ &= I_0 |(1+\cos \Psi + j \sin \Psi)^N| \\ &= I_0 |(\sqrt{1+2\cos \Psi + \cos^2 \Psi + \sin^2 \Psi})^N| = I_0 |(2+2\cos \Psi)^N| \\ &= 4 I_0 (1+\cos \Psi)^2 \end{aligned}$$

$$|AF|_{\max} = 4 I_0 (1+1)^2 = 16 I_0$$

$$|AF|_{\text{norm}} = \frac{|AF|}{|AF|_{\max}} = \frac{4 I_0 (1+\cos \Psi)^2}{16 I_0} = \frac{(1+\cos \Psi)^2}{4}$$

At 3dB points, $|AF|_{\text{norm}} = 0.707$

$$\frac{(1+\cos \Psi)^2}{4} = 0.707 \Rightarrow \cos \Psi = 0.681$$

$$\Psi = \pm 47.078^\circ$$

$$\Psi = \pi \cos \theta, \quad \alpha = 0, \quad d = 0.5\lambda$$

$$\Psi = Kd \cos \theta + \alpha$$

$$\cos \theta = \frac{\Psi}{\pi} = \frac{47}{180} \text{ or } \frac{-47}{180} \Rightarrow \theta = 74.86^\circ \text{ or } 105.14^\circ$$

$$3\text{dB beamwidth} = 105.14^\circ - 74.86^\circ = 30.28^\circ$$

Q. For $d = 0.25\lambda$, plot $|AF|_n$ for 2-element end-fire array

A. $d = 0.25\lambda$

$$0 \leq m \leq \frac{2d}{\lambda} \Rightarrow 0 \leq m \leq \frac{2 \times 0.25\lambda}{\lambda} \Rightarrow 0 \leq m \leq 0.5 \Rightarrow m = 0$$

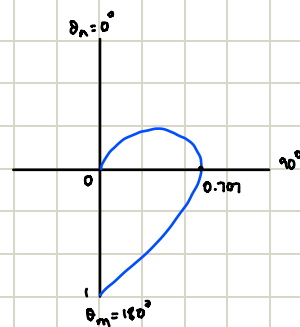
$$\cos \theta_m = \frac{m\lambda}{d} - 1 = 0 - 1 = -1 \Rightarrow \theta_m = \cos^{-1}(-1) = 180^\circ$$

$$n \leq \frac{2d}{\lambda} + \frac{1}{2} \Rightarrow n \leq \frac{2 \times 0.25\lambda}{\lambda} + \frac{1}{2} \Rightarrow n \leq 1 \Rightarrow n = 1$$

$$\cos \theta_n = \frac{2n-1}{2} \cdot \frac{\lambda}{d} - 1 = \frac{1}{2} \cdot \frac{\lambda}{0.25\lambda} - 1 = 1 \Rightarrow \theta_n = \cos^{-1}(1) = 0^\circ$$

$$AF = \cos \left[\frac{kd}{2} (1 + \cos \theta) \right] \quad (\theta = 90^\circ)$$

$$= \cos \left[\frac{2\pi}{\lambda} \cdot \frac{0.25\lambda}{2} (1 + \cos 90^\circ) \right] = \cos \left(\frac{\pi}{4} \times 1 \right) = \frac{1}{\sqrt{2}} = 0.707$$



Q. What is the maximum element spacing allowed in a binomial array so that no part of grating lobe appears in the visible region if it is a) Broadside array b) Endfire array

A. a) In binomial array, all zeros placed at $\psi = 180^\circ$

For broadside array, visible region can extend from $\psi = \pm 180$ through 0 to -180

So that no part of grating lobes appears in VR $\Rightarrow \psi = kd \cos \theta + \psi$

Max at $\psi = 0 \Rightarrow \theta = 90^\circ$

$\psi = 180^\circ \Rightarrow \theta = 0^\circ$

$$\pi = kd \Rightarrow d = \frac{\pi}{k} = \frac{\pi}{2\pi/\lambda} = \frac{\lambda}{2}$$

b) For endfire array, $\alpha = -kd$ & VR extends from $\psi = 0$ to $\psi = -\pi$

$$\psi = kd \cos \theta + \alpha$$

$$-\pi = -kd - kd = -2kd$$

$$\pi = 2 \cdot \frac{2\pi}{\lambda} \cdot d \Rightarrow d = \frac{\lambda}{4}$$

Q. How many grating lobes are present in the visible region of the array factor of a uniform array having an inter element spacing of 3λ and a uniform progressive phase shift $\alpha = -\frac{\pi}{3}$ in the excitation?

Sol:

$$\psi = kd \cos \theta + \alpha$$

Visible region extends from $+kd + \alpha$ to $-kd + \alpha$

$$\theta = 0^\circ \quad d = 3\lambda, \quad \alpha = -\frac{\pi}{3}$$

$$kd + \alpha = \frac{2\pi}{\lambda} \times 3\lambda + \left(-\frac{\pi}{3}\right) = 6\pi - \frac{\pi}{3}$$

$$\theta = \pi \quad -kd + \alpha = -\frac{2\pi}{\lambda} \times 3\lambda + \left(-\frac{\pi}{3}\right) = -6\pi - \frac{\pi}{3}$$

Grating lobes occur at $\psi = \pm 2m\pi$

$$m = 1, 2, 3, \dots$$

$$m = 1$$

$$\psi = \pm 2\pi$$

$$m = 2$$

$$\psi = \pm 4\pi$$

$$m = 3; \quad \psi = \pm 6\pi$$

$\psi = +6\pi$ which is not valid as visible region extends between $6\pi - \frac{\pi}{3}$ & $-6\pi - \frac{\pi}{3}$,

But $\psi = -6\pi$ can be considered.

So grating lobes are along

$$\psi = 4\pi, 2\pi, -2\pi, -4\pi, -6\pi$$

$\therefore$ There are 5 grating lobes.

$\therefore$ There are 5 grating lobes.

Q: Calculate the extent of the visible region of a uniform array along z-axis with $d = 2\lambda$ and $\alpha = \frac{\pi}{4}$. How many grating lobes are present in the visible region? What are the directions (in θ) of the main lobe and grating lobes?

Sol:

$$\psi = kd \cos \theta + \alpha$$

ψ varies from $kd + \alpha$ through α to $-kd + \alpha$

$$d = 2\lambda \quad \text{and} \quad \alpha = \frac{\pi}{4}$$

$$\therefore kd + \alpha = \left(\frac{2\pi}{\lambda} \times 2\lambda\right) + \frac{\pi}{4} = 4\pi + \frac{\pi}{4} = \frac{17\pi}{4}$$

$$-kd + \alpha = \left(-\frac{2\pi}{\lambda} \times 2\lambda\right) + \frac{\pi}{4} = -4\pi + \frac{\pi}{4} = -\frac{15\pi}{4}$$

The main lobe is along $\psi = 0 = kd \cos \theta + \alpha$
(principal maxima)

$$0 = kd \cos \theta + \alpha$$

$$0 = 4\pi \cos \theta + \frac{\pi}{4}$$

$$4\pi \cos \theta = -\frac{\pi}{4}$$

$$\cos \theta = -\frac{\pi}{4} \times \frac{1}{4\pi} = -\frac{1}{16}$$

$$\theta = \cos^{-1}\left(-\frac{1}{16}\right) = 93.58^\circ$$

grating lobes occur at $\psi = \pm 2m\pi$, $m = 1, 2, 3, \dots$

$$m = 1 \Rightarrow \psi = \pm 2\pi$$

$$m = 2 \Rightarrow \psi = \pm 4\pi$$

but -4π cannot be valid as $-kd + \alpha$ is $-4\pi + \frac{\pi}{4}$

$$\therefore \psi = +4\pi, +2\pi, -2\pi$$

$\therefore$ No. of grating lobes = 3

$$\psi = 4\pi \cos \theta + \frac{\pi}{4}$$

$$\psi = -2\pi$$

$$-2\pi = 4\pi \cos \theta + \frac{\pi}{4}$$

$$-2\pi - \frac{\pi}{4} = 4\pi \cos \theta$$

$$-\frac{8\pi - \pi}{4} = 4\pi \cos \theta$$

$$-\frac{7\pi}{4} = 4\pi \cos \theta$$

$$\cos \theta = -\frac{7\pi}{4 \times 4\pi}$$

$$\cos \theta = -\frac{7}{16}$$

$$\theta_1 = \cos^{-1}\left(-\frac{7}{16}\right) = 124.23^\circ$$

$$\theta_1 = 124.23^\circ$$

$$\psi = 2\pi$$

$$2\pi = 4\pi \cos \theta + \frac{\pi}{4}$$

$$\theta_2 = 64.06^\circ$$

$$\psi = +4\pi$$

$$4\pi = 4\pi \cos \theta + \frac{\pi}{4}$$

$$\theta_3 = 20.36^\circ$$